☠️ Code Path Mortality Tracker (CPMT)

Language: ANSI C
Architecture: Runtime Instrumentation & Failure Analytics System
Storage: Log-Based Execution Persistence
Version: 1.0

📌 Executive Summary

The Code Path Mortality Tracker (CPMT) is a runtime execution monitoring system designed to measure and analyze failure frequency across software execution paths.

The system integrates:

• Execution path tracking
• Failure detection
• Mortality rate computation
• Statistical evaluation
• Timestamp-based execution logging

CPMT simulates structured runtime instrumentation using deterministic execution counters and log persistence without external dependencies.

🏗️ System Architecture

CPMT follows a layered logical architecture:

• Execution Monitoring Layer
• Failure Detection Module
• Mortality Analytics Engine
• Log Persistence Layer

Although lightweight, the system models real-world observability patterns used in distributed systems and reliability engineering.

🔍 Execution Monitoring Layer

Each code path record contains:

• Path Name
• Total Executions
• Failure Count

Functional Capabilities:

• Track how many times a path executes
• Track how many times it fails
• Maintain runtime state

This simulates real production monitoring systems.

📊 Mortality Analytics Engine

The mortality rate measures failure density within a specific execution path.

Formula Used:

```
Mortality Rate = (Failures / Executions) × 100
```

Purpose:

• Identify unstable execution branches
• Quantify reliability
• Support debugging prioritization
• Simulate fault-tolerance analysis

📝 Log Persistence Layer

All execution events are recorded in:

```
cpmt_logs.txt
```

Logged Events:

• Path success
• Path failure

Each log entry includes timestamp metadata, simulating production-level trace logging.

🧠 Engineering Highlights

• Runtime execution tracking
• Failure frequency measurement
• Lightweight reliability modeling
• Log-based persistence
• Deterministic memory usage

🎯 Project Significance

CPMT demonstrates principles of:

• Software reliability engineering
• Runtime observability
• Failure rate modeling
• Instrumentation design

It reflects how modern monitoring systems evaluate execution stability.

🏁 Conclusion

The Code Path Mortality Tracker (CPMT) is a minimal yet conceptually powerful runtime reliability analysis tool implemented in pure ANSI C.

It integrates:

Execution Monitoring
Failure Detection
Mortality Computation
Audit Logging

All within a structured deterministic framework.